Data Science & Artificial Intelligence

Machine Learning

Nearest Neighbour

Discussion Notes

DPP-1

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[MCQ]

#Q. In k-NN, what is the effect of using weighted voting for classification?

A. It gives equal importance to all neighbors

B. It assigns more importance to closer neighbors

C. It reduces the effect of outliers

D. It increases the model's bias

* neighbors that are closer to the point being classified have a greater influence on the final classification decision than those are farther

→ Weighted Voting –

→ Closer neighbors

[MCQ]

#Q. What can be a consequence of using k-NN on a dataset with imbalanced classes?

A. Increased accuracy

B. Decreased accuracy

C. Balanced predictions

D. Improved precision

Majority of class more prevalent, KNN to predict the majority of class, ignoring minority of class.

* Imbalanced classes

[MCQ]

#Q. Which of the following scenarios is least suitable for k-NN?

A. Small datasets

B. Large datasets

C. High-dimensional datasets

D. Datasets with continuous target variables Answer: C. High-dimensional datasets

→ Least suitable for high dimensional data set – distance b/w data points become less meaningful.

[MCQ]

#Q. What is the primary challenge of using k-NN in high-dimensional spaces?

A. Overfitting

B. Underfitting

C. Curse of dimensionality

D. Scalability

Distance

Density of data points decrease exponentially as the number of dimension increase.

→ Difficulty in nearest neighbor Identification

↳ sparse.

[MCQ]

#Q. In k-NN regression, how is the final prediction made?

A. By taking the most frequent label

B. By taking the mean of the nearest neighbors' target values

C. By taking the median of the nearest neighbors' target values

D. By taking the mode of the nearest neighbors' target values

↳ Mean of the nearest neighbors target value.

↳ (Euclidean distance)

[MCQ]

#Q. Which technique can be used to reduce the dimensionality of the dataset before applying k-NN?

A. k-means clustering

B. PCA (Principal Component Analysis)

C. Linear Regression

D. Decision Trees

It transform the original variable into new set of orthogonal variables

* Maximum variance the data, allow, for ~~reducing~~ reduction in dimension

[MCQ]

#Q. What is the primary disadvantage of using the Manhattan distance in k-NN?

A. It is computationally expensive

B. It is sensitive to outliers

C. It can only be used for continuous data

D. It does not consider diagonal distance

↳ Does not consider diagonal distance.

→ horizontal & vertical component distance between point

[MCQ]

#Q. How can the performance of a k-NN classifier be evaluated?

A. By checking the training accuracy

B. By checking the computational time

C. By using a confusion matrix and calculating metrics such as precision, recall, and F1-score

D. By the number of neighbors

Confusion Matrix provides a detail break down of correct & incorrect prediction of the classifier

THANK - YOU